

Toy Example Illustration. Fig.1 illustrates a toy PPI network containing a drug d_1 targeting proteins s_1 and s_2 , and a disease characterized by target proteins t_1 and t_2 . Disease proteins weights are first computed, followed by weighted path evaluation for drug target protein and final drug score calculation.

Step1: Weighting disease target proteins.

$$w_t(t_1) = ppi_2 + ppi_3 + ppi_4 + ppi_6$$

$$w_t(t_2) = ppi_8$$

Step2: Weighting drug target proteins.

$$w_s(s_1) = (ppi_1 * ppi_2 + ppi_1 * ppi_5 * ppi_6) * (1 + w_t(t_1))$$

$$w_s(s_2) = (ppi_7 * ppi_8) * (1 + w_t(t_2))$$

Step3: Computing the Drug Score.

$$w_d(d_1) = dgi_1 * w_s(s_1) + dgi_2 * w_s(s_2)$$

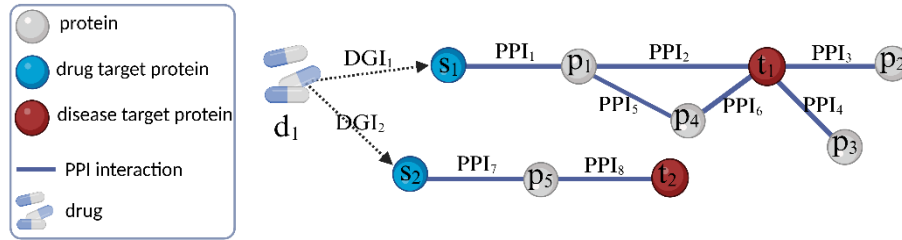


Fig. 1. A toy PPI network with a sample drug d_1 with targeted proteins s_1 and s_2 . And a disease sample with target proteins t_1 and t_2 .